

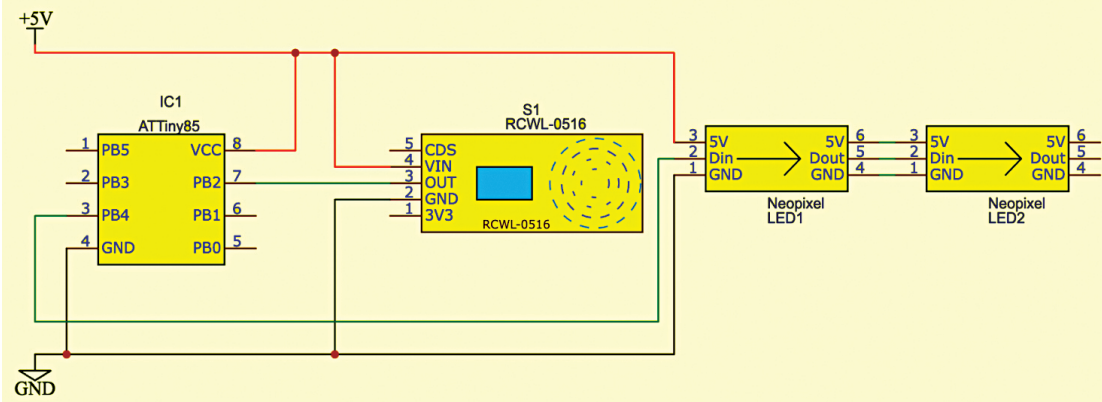


Fig. 2: Circuit diagram

sketch_jun17b.ino

```

1  #include <Adafruit_NeoPixel.h>
2
3  #define PIN_NEOPixel 4 // NeoPixel Data Pin
4  #define NUM_LEDS 60 // LED count
5  #define PIR_PIN 2 // PIR Sensor Pin
6
7  Adafruit_NeoPixel strip(NUM_LEDS, PIN_NEOPixel, NEO_GRB + NEO_KHZ800);
8
9  bool isLedOn = false;
10 unsigned long lastMotionTime = 0;
11 const unsigned long motionTimeout = 10000; // 10 seconds no motion = turn off
12 unsigned long lastColorChange = 0;
13 const unsigned long colorCycleDelay = 1000; // Change color every 1 second
14 int currentIndex = 0;
15
16 uint32_t colors[] = {
17   strip.Color(255, 244, 229), // Warm White
18   strip.Color(255, 99, 71), // Tomato (Warm Red)
19   strip.Color(255, 165, 0), // Orange
20   strip.Color(255, 215, 0), // Gold
21   strip.Color(240, 128, 128), // Light Coral
22   strip.Color(255, 182, 193), // Light Pink
23   strip.Color(255, 160, 122), // Light Salmon
24   strip.Color(218, 112, 214), // Orchid
25   strip.Color(255, 105, 180) // Hot Pink
26 };
27 };
28
29 void setup() {
30   pinMode(PIR_PIN, INPUT);
31   strip.begin();
32   strip.show(); // Clear LEDs
33 }
34
35 void loop() {

```

Fig. 3: Snippet of the source code

PIR motion sensor. However, both sensors function well according to their respective pin diagrams. The radar sensor is preferred in this controller because it can detect motion

through walls, plastics, and other non-metallic materials, while also providing a faster reaction time.

Once the circuit is powered on (as shown in Fig. 2), the RCWL-0516

sensor detects motion using its 3.2GHz microwave signal. Upon detecting an interruption, it keeps its output pin high for approximately 2 seconds. With continuous motion, the output

remains high, keeping the system active. The ATtiny85 microcontroller receives the input at PB2 (pin 7), and, according to the programmed logic, the output is provided through PB4 (pin 3) of the ATtiny85.

Software

The ATtiny85 is a microcontroller from the Atmel family, and it can be programmed using the Arduino board through the in-system programming (ISP) method. To programme it, an Arduino board with a USB cable, some male-to-male jumper wires, a 10µF capacitor, and a breadboard are required for pre-programming the microcontroller before placing it into the main circuit board.

To program the ATtiny85 IC, place the 8-pin ATtiny85 on the breadboard and make the following connections:

- Connect VCC of the Arduino to pin 8 (physical) of the ATtiny85
- Connect Arduino pin 13 to ATtiny85 pin 7
- Connect Arduino pin 12 to ATtiny85 pin 6
- Connect Arduino pin 11 to ATtiny85 pin 5